The Institution of Engineering and Technology

Member Transfer Form

Membership No. 83137730

If you require assistance or advice please telephone **+44 (0)1438 765678** or visit **www.theiet.org/contactmembership**. Please send the completed application form to: **Member Assessment Team, The IET, Michael Faraday House, Six Hills Way, Stevenage, Herts SG1 2AY UK**.

PLEASE USE BLACK INK AND WRITE CLEARLY IN THE BOXES IN BLOCK CAPITALS.
ALL FIELDS ARE MANDATORY. PLEASE COMPLETE ALL FIELDS TO AVOID ANY DELAY IN PROCESSING YOUR APPLICATION.

Your Details

Personal Details

Title: Mr
First Name(s): Philip Peter Andrew
Surname/Family Name: McGaw
Date of Birth: 01/10/1983 y y y y
Home Address: 28 Oakcroft Road
Birmingham
B13 0HP
United Kingdom
County/State:
Postcode:
Country:
Home Telephone: 07969 502 077
Personal Mobile:
Home Email Address: philip@mcgaw.eu

Employer Details

Job Title: Electrical Systems Engineer
Employer's Name: Jaguar Land Rover
Work Address: Jaguar Land Rover
B523/GO7
Gaydon
CV35 0RR
County/State:
Postcode:
Country:
Work Telephone:
Work Mobile:
Work Email Address: pmcgaw@jaguarlandrover.com

Preferred Name for Membership Card e.g. Mr A Jones: Mx Philip McGaw
Web username (if applicable):

Please specify the address that you would like all correspondence sent to: X Home ☐ Work

Please specify your gender: X Male ☐ Female

Data Protection

I understand and consent to the information provided on this form being processed by the IET for its sole use and that of its associated organisations, including my Local Network, for the purpose of promoting, delivering and improving my experience of the IET and its product and services or such other purposes as are described in the IET Privacy Statement. If either now or in the future I am based outside the European Economic Area (the "EEA"), my information may be transferred outside the EEA (for example to my Local Network or IET Global Office) to enable me to benefit from the IET opportunities overseas or, where required, to enable the IET to meet any legal or other legitimate obligations in that country. IET products and services are enjoyed by our members and customers globally. However if you do not wish to receive notification of products and services which may be of interest and benefit to you (by post or electronic means) please tick this box. ☐

Membership Details

Membership Category Options

To find out about the requirements for each category of membership visit **www.theiet.org/membership/types**

Member Application Section

Please refer to the guidance notes on the IET website **www.theiet.org/requirements** for further guidance in completing this section

X I wish to apply for **Member** with the designatory letters **MIET**

☐ I wish to apply for **Member** with the designatory letters **TMIET**

Qualifications

Please provide details of engineering or technology qualifications you hold or are in the process of completing in the table below:
(if you are applying for either the Advantage or Signature membership programme this section must be completed in full):

Award/Course e.g. BEng, MSc, NVQ3 etc	**Course Title** e.g. Electrical Engineering, Computing etc	**Name of University/College/Awarding Body and Country**	**Period of Study** (start and end date)

Work Experience

Depending on the Qualification that you hold we may need to assess your work experience to determine your eligibility for the level of membership selected. Following the assessment of your qualifications, we may contact you to provide a CV that clearly details your present employment position, previous employment positions, roles and responsibilities together with any other qualifications that are relevant to the interests of the IET.

Supporter's Details

Depending on the Qualification that you hold you may need a supporter. Your supporter can be your employer/line manager, a Member of the IET or someone that knows you in a work capacity e.g. client, previous employer etc. If you cannot find a suitable person to support your application, please contact the Membership Services Team on +44 (0)1438 765678.

☐ My supporter is not currently available to complete this section. Please contact me later to provide this information.

Title Mr First Name(s) Peter

Surname/Family Name Phillips Job Title Senior Manager - EMC & GRADE

X Line Manager X CEng ☐ IEng ☐ EngTech ☐ ICT*Tech* ☐ Associate ☐ TMIET X MIET ☐ FIET

Please detail membership of other Institutions e.g. MIMechE, MIEAust:

IET Membership No. (if applicable) Web username (if applicable)

Address for communication

County/State Postcode Country

Telephone Mobile

Email Address` pphilli3@jaguarlandrover.com

In my opinion this person has achieved the required knowledge and understanding and is a suitable candidate to become a Member of the IET and use the letters

☐ MIET* or ☐ TMIET* after their name (*please select as appropriate)

Signature of Supporter Date d d / m m / y y y y

The IET will contact your supporter for further background to your application for membership. It is not IET policy to disclose comments made by supporters and we would recommend applicants to discuss their membership application with the supporter prior to submission.

Declaration

I declare that the statements made on this form are to the best of my knowledge true. I agree to comply with the Royal Charter and Bye-laws and Rules of Conduct of the Institution of Engineering and Technology (IET) and understand that this is a commitment to behave ethically within my profession. I will do my best to promote the interests of the IET. I confirm that I have not committed any offence of which the IET would require me to give notice under its Rules of Conduct. The Rules of Conduct and the Royal Charter and Bye-laws are published on the website **www.theiet.org/byelaws** and the Terms and Conditions of Membership can be found **www.theiet.org/membership-terms**

Signature

Date 09/03/2020 y y y y

Over my work history I have acquired several technical skills such as advanced IT skills, Analogue Electronics with a view to RF suppression, and the use of Analogue instrumentation methods and how to correctly apply them when they are the most appropriate solution.

I have experience of working with test equipment such as Spectrum Analysers, Oscilloscopes, Electromagnetic Probes, GTEM Cells, and Digital Multi-meters to analyse and interpret the physical response of a circuit and evaluate against theoretical and simulated responses.

I am experienced at deriving theoretical circuit responses mentioned above from first principles and allying mathematical principals.

Some projects I have worked on required me to reverse engineer products without any design documentation to gain an insight into how the original designer came up with the solution and what modifications should be done to improve performance or reduce cost.

Once the product has been reverse engineered I was responsible for developing proposals such as improving the selection of components and addition of filtering to allow the device to fulfil its operational requirements and comply with EMC (Electromagnetic Compatibility) and LVD (Low Voltage Directive) directives.

This work was predominantly carried out alone, and without supervision. Research included taking notes, testing and analysing product performance, and reviewing datasheets and other technical specification documents. The use of test equipment was fundamental to this process, giving visibility on the inner workings of the circuits in the product.

Whilst working at CASS industries, I was responsible for training my co-works in the operation of the EMC test laboratory and how to perform the tests required to certify customers equipment. This also involved making them aware of the required legislation and standards that covered different tests and classes of equipment, the Testing and Reporting workload was split between myself and the other co-workers, in the case of more technical products I would take responsibility for these due to the challenges involved with testing and reporting on them.

There was a requirement to identify a situation where a Jig would improve the productivity, efficiency or simplify a process, or even allow us to offer a service that we had not got the facilities to offer before. Once the requirement was identified, I would be responsible for designing and building test jigs using both mechanical and electrical cad and using a mixture of rapid prototyping and machine shop tools; The combining of 3D printed parts, COTS (Common Off The Shelf) parts, and Electronics requires the ability to visualise how the parts will interface

While working at JLR as an EMC Component Engineering, I have significantly improved my project and people management skills as I regularly communicate with both JLR employees and external stakeholders via Email, phone, and face to face. This includes meetings where I am the only person representing the EMC department of JLR and ensuring compliance with the legal requirements, and JLR engineering standards.

Some of the requirements I'm responsible for ensuring compliance with have safety impact, requiring the correct operation of the component over the vehicle's entire life. One of the more problematic components I am working with is a 'Hands on Detection' ECU for a steering wheel, where I am collaborating with the component owner. I provide technical electrical guidance while he is talking to the manufacturer, while I also liaise with the testing facilities reporting the EMC failures and advise on component re-design to resolve the failed tests.

I have recently performed a research investigation into the applicability of GB 8702 and ICNIRP standards to JLRs line of EVs (electric vehicles). The outcome of this investigation was a presentation to my colleges with recommendations on EMC emission limits that should be applied by JLR to comply with the legal requirements in all markets. To support this work, I undertook an investigation on my own initiative to perform back to back comparison of three different types of Electromagnetic probes and justified my recommendation at the type most appropriate for supporting JLR in testing going forward. While there has been no formal change of test procedures yet, there is active discussion on what the best procedure is. I have been performing more investigation work to gain understanding about the issues around EMF and the implications regarding components within the vehicle. The longer-term output from this is to feed the understanding back into a revised version of the JLR component specification.

There is an on-gong project to build new EMC testing facilities within JLR, I was responsible for producing the technical requirements document for the two new labs, Amplifier room, and the auxiliary equipment. This document was used by three prospective suppliers to come up with technical proposals and quotes, and then was used as the basis for judging the submissions against the requirements. The outcome of this was that one supplier was awarded the work as they clearly managed to meet JLRs requirements and showed good value for money.

While at JLR I have continued to learn new skills, taking advantage of the companies training scheme to acquire more JLR qualifications that are relevant for my roll; I am also studying external courses on Coursera covering ADAS (Advanced driver-assistance systems), and other autonomous driving training, that are less focused on what I am currently doing in work, and more on where I would like to be in five years, working with more cutting edge technologies.

I hold the permits from JLR to work safely on 'Lid off Batteries', Live un-switched hazardous voltage, and de-power and lock off electric vehicles. I also have received training on how to perform formal and informal risk assessments, I can also authorise other staff to work on vehicles I have de-powered.

Outside work I have several interests, such as industrial heritage, robotics, rapid prototyping, and photography. Some of my personal projects include:

The Design and construction of an inexpensive PnP (pick and place) machine using common off the shelf aluminium extrusions profiles and parts of the body and mechanism are to be printed on an FDM (Fused Deposition Modelling) 3D printer. The 3D printed parts are

designed by me in Fusion 360, while the PCBs are designed using KiCAD, and hand assembled.

The aim is to make a device that is easy and inexpensive to replicate and allows for the £2 level of PCBs from china to be populated in a home workshop. The objective of this project is to be able to reliably and repeatably place devices down to 0603 size.

The Design and construction of a ROV (Remotely Operated Underwater Vehicle) that can be operated from a Laptop or iPad allowing the operator to explore underwater.

The main parts of the body are 3D printed on an FDM 3D printer. Parts of the pressure vessel are machined on a Laser Cutter, some of the smaller parts are printed using a DLP (Digital Light Processing) 3D printer.

Designing and development of a cheap educational robot platform that can be used to experiment with swarm behaviours, ML (Machine Learning) and AI (Artificial Intelligence)

This includes writing software and firmware in 'Arduino', Python and ROS (Robot Operating System).